STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-DG-HULL-ACYCLIC

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3572-3583 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3585-3676, the complete author proof (90 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_sdga.json (pb=3705, stmt 3572-3583, proof 3585-3676).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSK: sources/stacks-sdga/pinned-original-source.tex, source lines 3513–3520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRD: sources/stacks-sdga/pinned-original-source.tex, source lines 1253–1257. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FS8: sources/stacks-sdga/pinned-original-source.tex, source lines 2785–2804. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRT: sources/stacks-sdga/pinned-original-source.tex, source lines 2137–2225. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FR7: sources/stacks-sdga/pinned-original-source.tex, source lines 873–998. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRV: sources/stacks-sdga/pinned-original-source.tex, source lines 2248–2327. Cited by the retained statement or proof but not part of

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FR9: sources/stacks-sdga/pinned-original-source.tex, source lines 1014–1097. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The functors F, G of Lemma 0FSK have the following properties. Given a graded $\mathcal{A}$-module $\mathcal{N}$ we have*

(1) *the counit $\mathcal{N} \to F(G(\mathcal{N}))$ is injective,*
(2) *the map $\overline{\mathrm{d}} : \mathcal{N} \to \mathrm{Coker}(\mathcal{N} \to F(G(\mathcal{N})))[1]$ is an isomorphism, and*
(3) *$G(\mathcal{N})$ is an acyclic differential graded $\mathcal{A}$-module.*

Proof. We observe that property (3) is a consequence of properties (1) and (2). Namely, if s is a nonzero local section of $F(G(\mathcal{N}))$ with $\mathrm{d}(s) = 0$, then s cannot be in the image of $\mathcal{N} \to F(G(\mathcal{N}))$. Hence we can write the image $\overline{s}$ of s in the cokernel as $\overline{\mathrm{d}}(s')$ for some local section s' of $\mathcal{N}$. Then we see that $s = \mathrm{d}(s')$ because the difference $s - \mathrm{d}(s')$ is still in the kernel of d and is contained in the image of the counit.

Let us write temporarily $\mathcal{A}_{gr}$, respectively $\mathcal{A}_{dg}$ the sheaf $\mathcal{A}$ viewed as a (right) graded module over itself, respectively as a (right) differential graded module over itself. The most important case of the lemma is to understand what is $G(\mathcal{A}_{gr})$. Of course $G(\mathcal{A}_{gr})$ is the object of $Mod(\mathcal{A}, \mathrm{d})$ representing the functor

$$\mathcal{M} \longmapsto \mathrm{Hom}_{Mod(\mathcal{A})}(\mathcal{A}_{gr}, F(\mathcal{M})) = \Gamma(\mathcal{C}, \mathcal{M})$$

By Remark 0FS8 we see that this functor represented by $C[-1]$ where C is the cone on the identity of $\mathcal{A}_{dg}$. We have a short exact sequence

$$0 \to \mathcal{A}_{dg}[-1] \to C[-1] \to \mathcal{A}_{dg} \to 0$$

in $Mod(\mathcal{A}, \mathrm{d})$ which is split by the counit $\mathcal{A}_{gr} \to F(C[-1])$ in $Mod(\mathcal{A})$. Thus $G(\mathcal{A}_{gr})$ satisfies properties (1) and (2).

Let U be an object of $\mathcal{C}$. Denote $j_U : \mathcal{C}/U \to \mathcal{C}$ the localization morphism. Denote $\mathcal{A}_U$ the restriction of $\mathcal{A}$ to U. We will use the notation $\mathcal{A}_{U,gr}$ to denote $\mathcal{A}_U$ viewed as a graded $\mathcal{A}_U$-module. Denote $F_U : Mod(\mathcal{A}_U, \mathrm{d}) \to Mod(\mathcal{A}_U)$ the forgetful functor and denote G_U its adjoint. Then we have the commutative diagrams

$$\begin{array}{ccc} Mod(\mathcal{A}, \mathrm{d}) & \xrightarrow{F} & Mod(\mathcal{A}) \\ {\scriptstyle j_U^*}\downarrow & & \downarrow{\scriptstyle j_U^*} \\ Mod(\mathcal{A}_U, \mathrm{d}) & \xrightarrow{F_U} & Mod(\mathcal{A}_U) \end{array} \quad \text{and} \quad \begin{array}{ccc} Mod(\mathcal{A}_U, \mathrm{d}) & \xrightarrow{F_U} & Mod(\mathcal{A}_U) \\ {\scriptstyle j_{U!}}\downarrow & & \downarrow{\scriptstyle j_{U!}} \\ Mod(\mathcal{A}, \mathrm{d}) & \xrightarrow{F} & Mod(\mathcal{A}) \end{array}$$

by the construction of j_U^* and $j_{U!}$ in Sections 0FR7, 0FRT, 0FR9, and 0FRV. By uniqueness of adjoints we obtain $j_{U!} \circ G_U = G \circ j_{U!}$. Since $j_{U!}$ is an exact functor, we see that the properties (1) and (2) for the counit $\mathcal{A}_{U,gr} \to F_U(G_U(\mathcal{A}_{U,gr}))$ which we've seen in the previous part of the proof imply properties (1) and (2) for the counit $j_{U!}\mathcal{A}_{U,gr} \to F(G(j_{U!}\mathcal{A}_{U,gr})) = j_{U!}F_U(G_U(\mathcal{A}_{U,gr}))$.

In the proof of Lemma 0FRD we have seen that any object of $Mod(\mathcal{A})$ is a quotient of a direct sum of copies of $j_{U!}\mathcal{A}_{U,gr}$. Since G is a left adjoint, we see that G commutes with direct sums. Thus properties (1) and (2) hold for direct sums of objects for which they hold. Thus we see that every object $\mathcal{N}$ of $Mod(\mathcal{A})$ fits into an exact sequence

$$\mathcal{N}_1 \to \mathcal{N}_0 \to \mathcal{N} \to 0$$

such that (1) and (2) hold for $\mathcal{N}_1$ and $\mathcal{N}_0$. We leave it to the reader to deduce (1) and (2) for $\mathcal{N}$ using that G is right exact. $\square$